

Qiuqi Wang, Ph.D., A.S.A.

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Academic Appointment

Georgia State University
– Assistant Professor, Maurice R. Greenberg School of Risk Science
08/2023 – present

Education

Ph.D. Actuarial Science, University of Waterloo
– Advisor: Ruodu Wang
06/2023
M.Phil. Mathematics, Hong Kong University of Science and Technology
– Advisor: Yue Kuen Kwok
07/2019
B.S. Mathematics and Economics, Hong Kong University of Science and Technology
07/2017
International Exchange Student, Technical University of Munich
04/2016 – 09/2016

Professional Designation

Associate of the Society of Actuaries (A.S.A.)
08/2023

Editorial Roles

Co-Guest Editor
International Journal of Theoretical and Applied Finance
Special Issue on “Risk Measures, Model Uncertainty, and Robust Financial Decision-Making”
2026-2027

Research Interests

- Actuarial science; quantitative risk management
- Operations research; financial engineering
- Statistics; anytime valid testing, e-values
- Economics; game theory

Publications and Manuscripts

Pre-publication Manuscripts

- [1] Jiao, Z., **Wang, Q.** and Zhao, Y. (2025). Comparative e-backtests for general risk measures. Preprint.
- [2] Bellini, F., Huang, M., **Wang, Q.** and Wang, R. (2025). Lambda Expected Shortfall. Preprint.
- [3] Chi, Y., Fung, T. C., Peng, L. and **Wang, Q.** (2025). Statistical solutions of optimal multivariate reinsurance under distortion risk measures. Preprint.

Peer-reviewed Journal Articles

- [1] **Wang, Q.**, Wang, R. and Ziegel, J. (2026). E-backtesting. *Management Science*, **72**(6), 4952–4973.
- [2] Han, X., **Wang, Q.**, Wang, R. and Xia, J. (2026). Cash-subadditive risk measures without quasi-convexity. *Mathematics of Operations Research*, **51**(2), 1205–1226.
- [3] Boonen, T. J., Chen, Y., Han, X. and **Wang, Q.** (2025). Optimal insurance design with Lambda-Value-at-Risk. *European Journal of Operational Research*, **327**(1), 232–246.
- [4] Aboagye, E., Asimit, V., Fung, T. C., Peng, L. and **Wang, Q.** (2025). A revisit of the optimal excess-of-loss contract. *European Journal of Operational Research*, **322**(1), 341–354.
- [5] Pesenti, S. M., **Wang, Q.** and Wang, R. (2025). Optimizing distortion riskmetrics with distributional uncertainty. *Mathematical Programming*, **213**(1-2), 51–106.
- [6] **Wang, Q.**, Wang, R. and Zitikis, R. (2022). Risk measures induced by efficient insurance contracts. *Insurance: Mathematics and Economics*, **103**, 56–65.
- [7] Embrechts, P., Mao, T., **Wang, Q.** and Wang, R. (2021). Bayes risk, elicibility, and the Expected Shortfall. *Mathematical Finance*, **31**(4), 1190–1217.
- [8] **Wang, Q.**, Wang, R. and Wei, Y. (2020). Distortion riskmetrics on general spaces. *ASTIN Bulletin*, **50**(3), 827–851.
- [9] **Wang, Q.** and Kwok, Y. K. (2020). Real option signaling games of debt financing using equity guarantee swaps under asymmetric information. *International Journal of Theoretical and Applied Finance*, **23**(05), 2050036.
- [10] **Wang, Q.** and Kwok, Y. K. (2019). Signaling game models of equity financing under information asymmetry and finite project life. *International Journal of Financial Engineering*, **06**(01), 1950002.

Dissertations

- [1] Wang, Q. (2023). Characterizing, optimizing and backtesting metrics of risk. *Ph.D. Thesis*. University of Waterloo, Canada.
- [2] Wang, Q. (2019). Real option signaling games under asymmetric information and finite option life. *M.Phil. Thesis*. Hong Kong University of Science and Technology, Hong Kong.

Selected Honors and Awards

3rd place of Mathematics Doctoral Prize, UW	2024
Statistical Society of Canada's Pierre Robillard Award, SSC	2024
<i>An award with one recipient per year recognizing the best Ph.D. thesis defended at a Canadian university in a given year in the fields of probability or statistics</i>	
James C. Hickman Scholar Doctoral Stipend, SOA	2021 – 2023
<i>An award with annually 3-5 recipients currently enrolled in or applying to actuarial science doctoral programs or a related field, in the U.S. or Canada, recognizing great academic performance and potential</i>	
Honorable Mention in ARC Graduate Student Presentation Awards, SOA	2022
SAS Chair's Award, UW	2021 – 2022
Comprehensive Award (out of 2 recipients), UW	2020
International Doctoral Student Award, UW	2019 – 2023
Statistics and Actuarial Science Doctoral Entrance Award, UW	2019
Postgraduate Studentship, HKUST	2017 – 2019
Scholarship Scheme for Continuing Undergraduate Students, HKUST	2015 – 2017
Fong Shu Chuen Scholarship, HKUST	2015 – 2016
School of Science Dean's List, HKUST	2014

Academic Visits

Monash University (Department of Econometrics and Business Statistics) (Clayton, Australia)	2026
University of Melbourne (Department of Economics) (Melbourne, Australia)	2026
University of New South Wales (School of Risk & Actuarial Studies) (Sydney, Australia)	2026
University of Iowa (Department of Statistics and Actuarial Science) (Iowa City, US)	2026
Michigan State University (Department of Statistics and Probability) (Lansing, US)	2025
University of Waterloo (Department of Statistics and Actuarial Science) (Waterloo, Canada)	2023, 2025
Nankai University (School of Mathematical Sciences) (Tianjin, China)	2023, 2025
Chinese University of Hong Kong-Shenzhen (School of Science and Engineering) (Shenzhen, China)	2024
Chinese Academy of Sciences (Academy of Math and Systems Science) (Beijing, China)	2021, 2023
University of Science and Technology of China (School of Management) (Hefei, China)	2021

Conferences Organized

- 3rd Waterloo Student Conference in Statistics, Actuarial Science and Finance (*Co-chair*), 2022 (Waterloo, Canada)

- 2nd Waterloo Student Conference in Statistics, Actuarial Science and Finance (*Co-chair*), 2021 (Online)

Invited Academic Presentations

- Institute of Mathematical Statistics - Asia Pacific Rim Meeting, 2026 (Hong Kong, China)
- 2nd Midwest Actuarial Research Seminars, 2026 (West Lafayette, US)
- China Conference on Insurance and Risk Management, 2025 (Urumqi, China)
- 3rd Foundations and Applications of Decentralized Risk-Sharing, 2025 (Paris, France)
- Young Talents in Actuarial Science and Quantitative Finance, 2025 (Waterloo, Canada)
- Pierre Robillard Invited Address, SSC Annual Meeting, 2024 (St. John's, Canada)
- Young Data Science Researcher Seminar Zurich, 2022 (Online)
- INFORMS Annual Meeting, 2022 (Indianapolis, US)

Contributed Academic Presentations

- 3rd Safe and Anytime Valid Inference (SAVI) Conference, 2026 (Enschede, Netherlands)
- 1st ASTIN Bulletin Conference, 2026 (Zurich, Switzerland)
- 60th Actuarial Research Conference, 2025 (Toronto, Canada)
- 59th Actuarial Research Conference, 2024 (Murfreesboro, US)
- 27th International Congress of Insurance: Mathematics and Economics, 2024 (Chicago, US)
- 57th Actuarial Research Conference, 2022 (Urbana-Champaign, US)
- 25th International Congress on Insurance: Mathematics and Economics, 2022 (Online)
- CORS/INFORMS International Conference, 2022 (Vancouver, Canada)
- 56th Actuarial Research Conference, 2021 (Online)
- 24th International Congress on Insurance: Mathematics and Economics, 2021 (Online)
- 5th PKU-NUS Annual International Conference on Quantitative Finance and Economics, 2021 (Online)
- 5th Asian Quantitative Finance Seminar, 2020 (Online)
- 1st Waterloo Student Conference in Statistics, Actuarial Science and Finance, 2019 (Waterloo, Canada)
- 23rd Annual International Real Options Conference, 2019 (London, UK)

Teaching Experience

Instructor – Georgia State University

AS 3230/8230 - Financial Mathematics

Spring 2024; Fall
2024–2026

RMI 8300 - Predictive Risk Modeling

Fall 2024, 2025

Instructor – University of Waterloo

ACTSC 371 - Introduction to Investment

Winter 2023

Teaching Assistant – University of Waterloo

ACTSC 363 - Casualty and Health Insurance Mathematics 1

Fall 2022

ACTSC 964 - Quantitative Risk Management

Spring 2022

ACTSC 446/846 - Mathematics of Financial Markets

Winter 2022

MATBUS 472 - Risk Management

Winter 2021–2022

MATBUS 470 - Derivatives

Fall 2019–2022;
Winter 2021

MATBUS 471 - Fixed Income Securities

Fall 2019–2021;
Winter 2020

ACTSC 431 - Loss Model I / Casualty and Health Insurance Mathematics 2

Spring 2020, 2021

ACTSC 445/845 - Quantitative Enterprise Risk Management

Spring 2020

ACTSC 231 - Introductory Financial Mathematics

Winter 2020

Teaching Assistant – HKUST

MATH 4321 - Game Theory (with tutorials)

Spring 2018, 2019

MATH 4511 - Quantitative Methods for Fixed Income Derivatives (with tutorials)

Fall 2018

Peer-review Service

- Actuarial: *Insurance: Mathematics and Economics*, *ASTIN Bulletin*, *North American Actuarial Journal*, *European Actuarial Journal*, *Annals of Actuarial Science*
- Finance: *Mathematical Finance*, *Finance & Stochastics*, *SIAM Journal on Financial Mathematics*, *Quantitative Finance*
- OR: *Operations Research*, *Mathematics of Operations Research*, *European Journal of Operational Research*
- Probability: *Statistics & Probability Letters*
- Economics: *Economic Modelling*